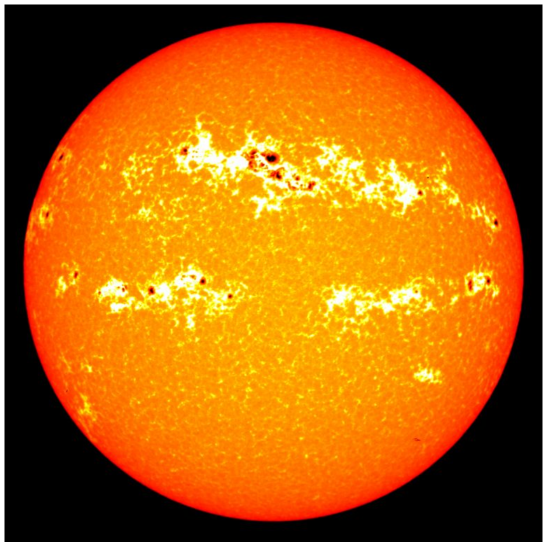
# Exercises



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ESBC-ENER751 Introduction to radiative transfer
Solar Academy
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*Freely adapted from the original content by Cyril Caliot.*

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## [Ex0] General knowledge

1. Compute the Stefan-Boltzmann constant $\sigma$ up to 7 significant digits.

2. Observed from the Earth's surface, the Sun holds a solid angle $\Omega_{S \to E}$ that varies throughout the year. Calculate the solid angles of the Sun at the Earth's Aphelion (July) and Perihelion (January) (Figure 1).

3. The total (spectrally integrated) solar irradiance at the top of atmosphere at a distance of one astronomical unit ($1.5.10^8$ km) is called the Solar Constant, and equals $H_S = 1361\,\mathrm{W.m^{-2}}$. Calculate its value at the Aphelion and Perihelion.

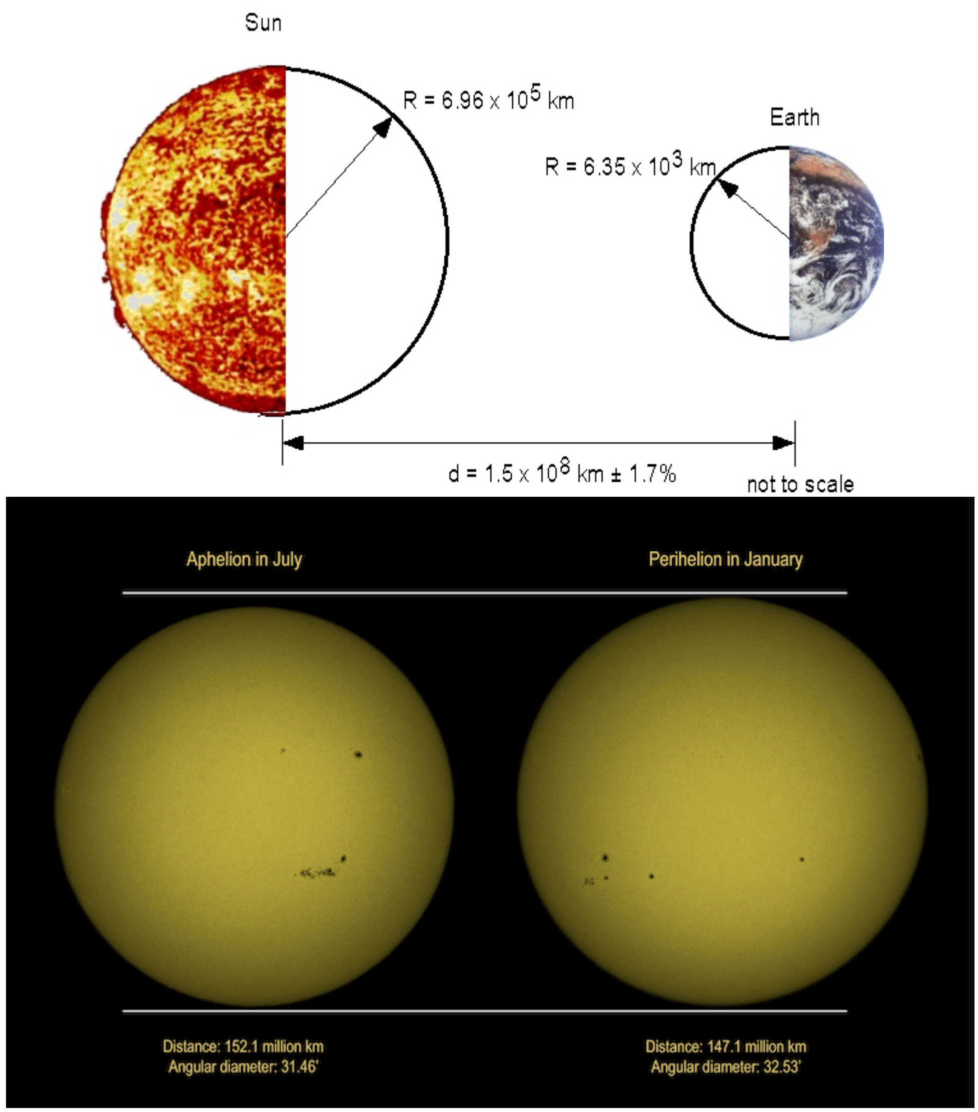


Figure 1 Distance between the Earth and the Sun at the Aphelion (~2 weeks after the summer solstice) and Perihelion (~2 weeks after the winter solstice).

## [Ex1] About blackbodies

1. Express the literal fraction of blackbody emission $F_{0\to n\lambda_{\max}T}$ between $\lambda = 0\,\mathrm{nm}$ and $\lambda_{\max}$, its wavelength of maximum emission, as a function of its temperature $T$ and main constants in $\sigma$. Then, give a simple formulation for $F_{0\to n\lambda_{\max}T}$, with respect to its spectral radiation function $f_\lambda$. Tabulated values for $f_\lambda$ are given in Figure 3.

2. What are the lower and upper wavelengths, where $\sim 96\,\%$ of radiation is emitted, for a blackbody at $20\,°\mathrm{C}$ and $5780\,\mathrm{K}$?

3. A Lambertian gray body of temperature $T = 500\,\mathrm{K}$ has an emittance $E = 2\,835\,\mathrm{W.m^{-2}}$. Give the value of its emissivity $\epsilon$. The same material is heated up and now emits $E' = 45\,360\,\mathrm{W.m^{-2}}$. What is the value of its new temperature $T'$?

4. Considering the different steps in the establishment of a spectral measurement (Figure 2), **find** an equation relating the blackbody emission of the object to the global spectral response, assuming that the atmosphere is not emitting (cold atmosphere).

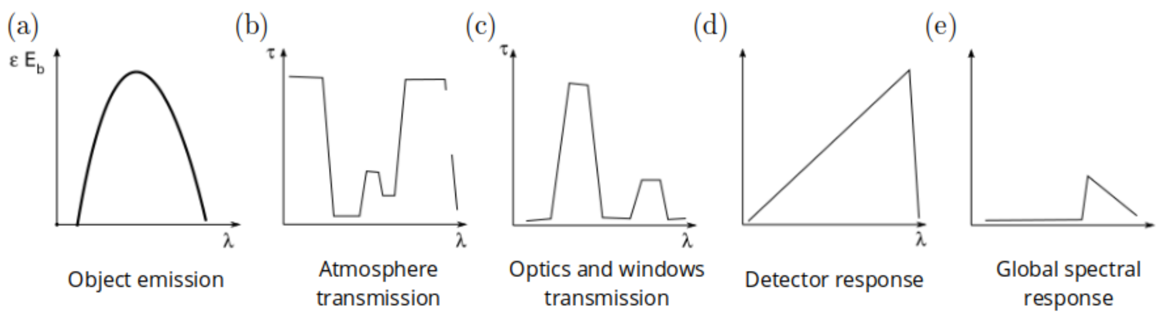


Figure 2 Measurement steps of a the spectral irradiance emitted by an object.

Blackbody radiation functions $f_\lambda$

| $\lambda T$, $\mu m \cdot K$ | $f_\lambda$ | $\lambda T$, $\mu m \cdot K$ | $f_\lambda$ |
|---|---|---|---|
| 200 | 0.000000 | 6200 | 0.754140 |
| 400 | 0.000000 | 6400 | 0.769234 |
| 600 | 0.000000 | 6600 | 0.783199 |
| 800 | 0.000016 | 6800 | 0.796129 |
| 1000 | 0.000321 | 7000 | 0.808109 |
| 1200 | 0.002134 | 7200 | 0.819217 |
| 1400 | 0.007790 | 7400 | 0.829527 |
| 1600 | 0.019718 | 7600 | 0.839102 |
| 1800 | 0.039341 | 7800 | 0.848005 |
| 2000 | 0.066728 | 8000 | 0.856288 |
| 2200 | 0.100888 | 8500 | 0.874608 |
| 2400 | 0.140256 | 9000 | 0.890029 |
| 2600 | 0.183120 | 9500 | 0.903085 |
| 2800 | 0.227897 | 10,000 | 0.914199 |
| 3000 | 0.273232 | 10,500 | 0.923710 |
| 3200 | 0.318102 | 11,000 | 0.931890 |
| 3400 | 0.361735 | 11,500 | 0.939959 |
| 3600 | 0.403607 | 12,000 | 0.945098 |
| 3800 | 0.443382 | 13,000 | 0.955139 |
| 4000 | 0.480877 | 14,000 | 0.962898 |
| 4200 | 0.516014 | 15,000 | 0.969981 |
| 4400 | 0.548796 | 16,000 | 0.973814 |
| 4600 | 0.579280 | 18,000 | 0.980860 |
| 4800 | 0.607559 | 20,000 | 0.985602 |
| 5000 | 0.633747 | 25,000 | 0.992215 |
| 5200 | 0.658970 | 30,000 | 0.995340 |
| 5400 | 0.680360 | 40,000 | 0.997967 |
| 5600 | 0.701046 | 50,000 | 0.998953 |
| 5800 | 0.720158 | 75,000 | 0.999713 |
| 6000 | 0.737818 | 100,000 | 0.999905 |

Figure 3 Tabulated values for the spectral blackbody radiation function $f_\lambda$.

## [Ex2] A selective opaque surface

An opaque surface presents a selective emissivity, with its value changing at $\lambda = 0.8\,\mu\text{m}$ (Figure 4). What is the average solar absorptivity of this surface?

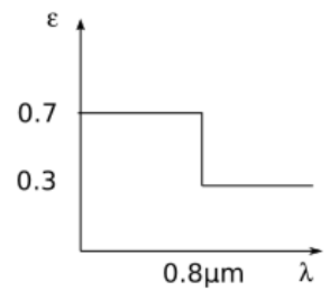


Figure 4 Emission spectrum of a selective opaque surface.

**[Ex3] Solid angles and view factors**

1. Show step by step that the infinitesimal solid angle $d\Omega$ is equal to:

$$d\Omega = \sin\theta \, d\theta \, d\varphi,$$

with $\theta$ and $\varphi$ the angular coordinates of the three-dimensional space. What is the value of the solid angle for the complete sphere? And for the hemisphere?

2. Show step by step that:

$$\int_{2\pi} \int_0^{\theta_1} \cos(\theta)\sin(\theta) \, d\theta \, d\varphi = \pi(1 - \cos^2\theta_1).$$

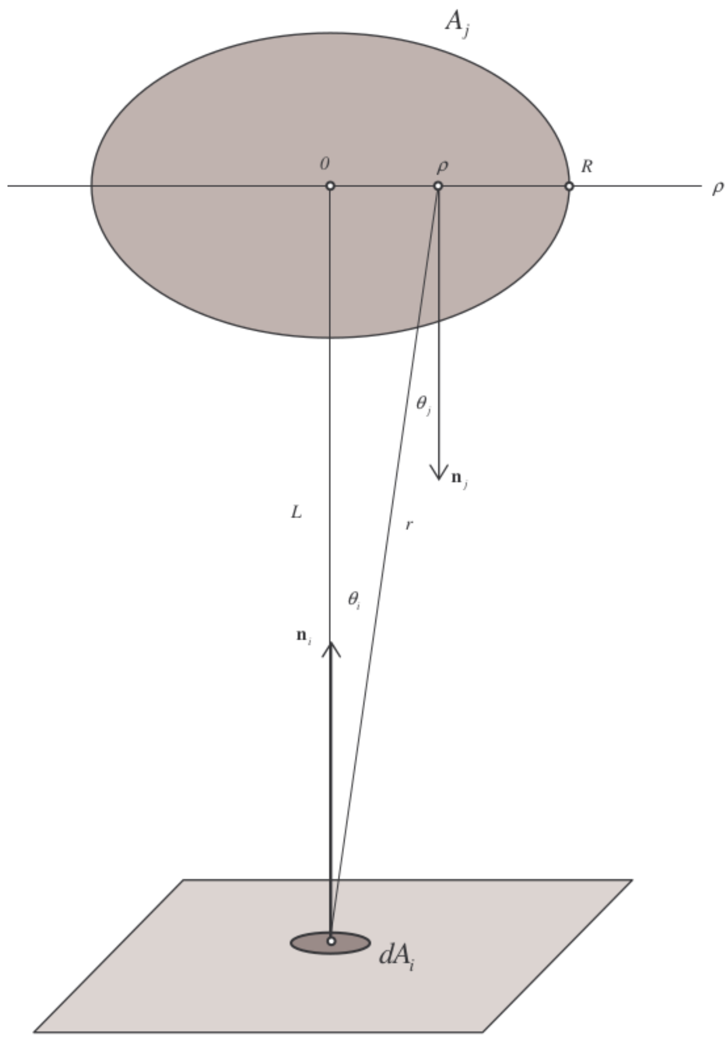


Figure 5 Elementary surface $dA_i$ facing a disk of area $A_j$.

3. We now consider an elementary surface $dA_i$ looking towards a disk of surface $A_j$

(Figure 5). Demonstrate that the view factor from $dA_i$ to $A_j$ is:

$$F_{dA_i \to A_j} = \frac{1}{1 + \left(\frac{L}{R}\right)^2}.$$

4. Determine the view factors $F_{1\to 2} \equiv F_{12}$ and $F_{2\to 1} \equiv F_{21}$ for the different configurations in Figure 6 using the reprocity theorem and other basic shape factor relations. Do not use tables or charts. What is the value of $F_{22}$ for configuration (b)? What about $F_{23}$ and $F_{32}$ for (f)?

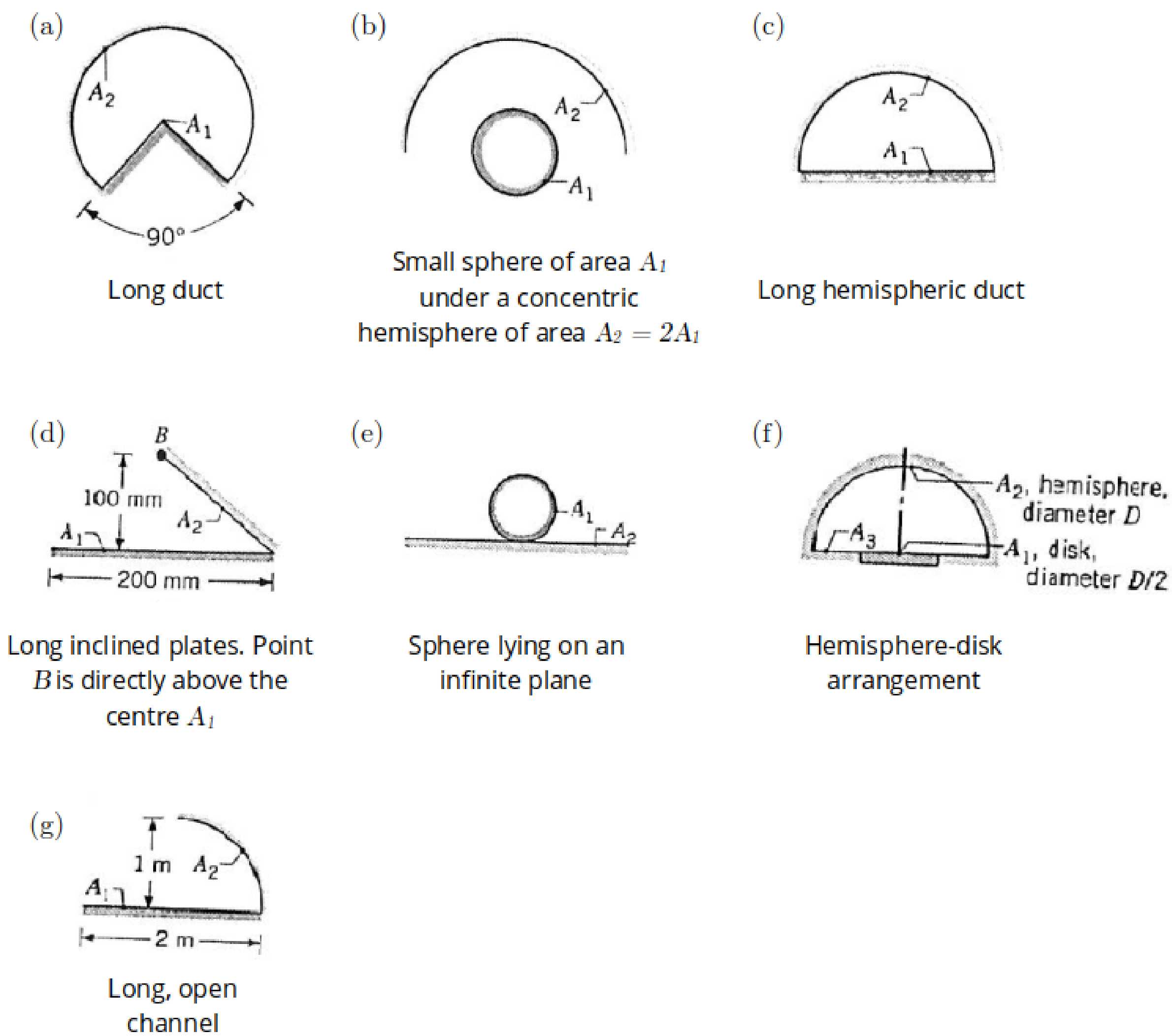


Figure 6 Simple configurations of facing objects of surfaces $A_1$ and $A_2$, respectively.

5. We finally consider a rectangular area $A_i$ facing another area $A_{(j)}$ being a discrete sum

of $n$ planar surfaces $a_k$, $k \in \{1, 2, \dots, n\}$, (Figure 7) so that:

$$A_{(j)} = a_1 + a_2 + \cdots + a_n.$$

Demonstrate the following equation:

$$F_{(j)i} = \frac{a_1 F_{1i} + a_2 F_{2i} + \cdots + a_n F_{ni}}{A_{(j)}},$$

with $F_{ik}$ and $F_{ki}$ the view factors from $A_i$ to $a_k$ and from $a_k$ to $A_i$, respectively.

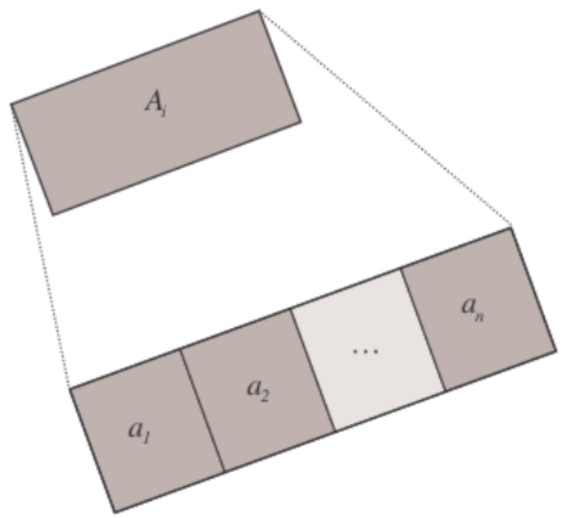


Figure 7 Rectangular surface $A_i$ facing a discrete sampled surface $A_{(j)} = a_1 + a_2 + \cdots + a_n$, $n \in \mathbb{N}_+^*$.

**[Ex4] Radiative transfers in a furnace cavity**

A furnace cavity, in the form of a 75 mm diameter and 150 mm length cylinder, is open at one end to large surroundings at temperature $T_{ext} = 27\,°C$ (Figure 8). The sides and bottom of the cavity may be assimilated to balckbodies. They are heated electrically, well insulated, and maintained at temperature $T_1 = 1\,350\,°C$ and $T_2 = 1\,650\,°C$, respectively.

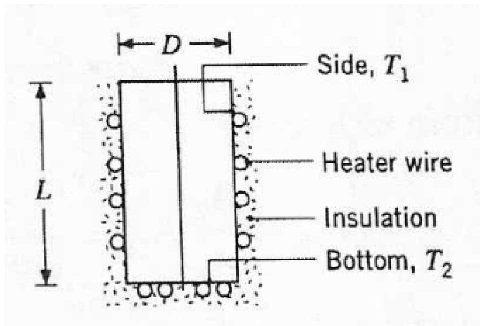


Figure 8 Cylindric furnace cavity.

How much power $P$ is required to maintain the furnace conditions?

## [Ex5] Radiative exchanges between a heater and a curved absorber

In manufacturing, the special coating on a curved solar absorber surface of area $A_2 = 15\,\mathrm{m}^2$ is cured by exposing it to an infrared heater of width $W = 1\,\mathrm{m}$ (Figure 9). Both absorber and heater are of length $L = 10\,\mathrm{m}$ and are separated by a distance $H = 1\,\mathrm{m}$. The heater is at temperature $T_1 = 1\,000\,\mathrm{K}$ and has an emissivity $\epsilon_1 = 0.9$, while the absorber is at temperature $T_2 = 600\,\mathrm{K}$ with an emissivity $\epsilon_2$ of 0.5. The system is located in a large room whose wall surfaces are at ambient temparture ($T_{sur} = 300\,\mathrm{K}$).

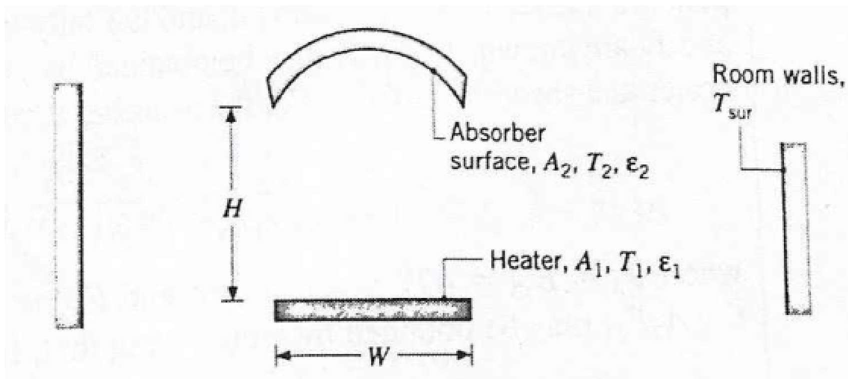


Figure 9 Absorber-heater system configuration.

What is the net rate of heat transfer to the absorber surface? Interpret its sign.

## [Ex6] Radiative transfers in a painted baking oven

A painted baking oven consists of a long, triangular duct in which a heated surface is maintained at $T_1 = 1\,200\,\mathrm{K}$ and another surface is insulated (temperature $T_R$) (Figure 10). Painted panels, which are maintained at temperature $T_2 = 500\,\mathrm{K}$, occupy the third surface. The triangle is of width $W = 1\,\mathrm{m}$ on a side, and the heated and insulated surfaces have an emissivity $\epsilon_1 = \epsilon_R = 0.8$. The panels emissivity $\epsilon_2$ is 0.4.

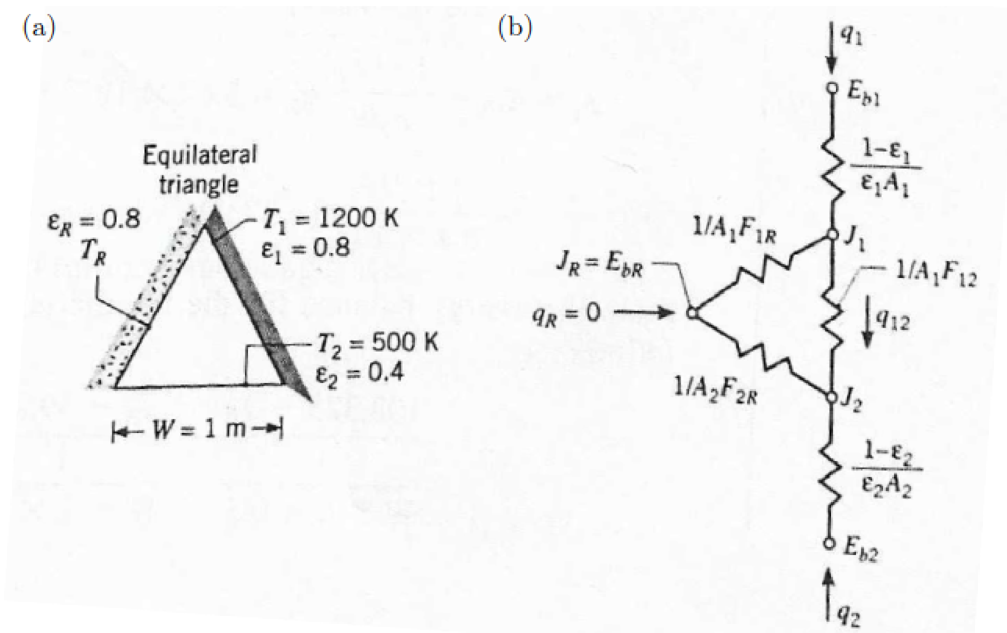


Figure 10 Schematic of the baking oven (a) and involved radiative exchanges (b).

During steady-state operation, at what rate must the energy be supplied to the heated side, per unit length of the duct, to maintain its temperature at $1\,200\,\mathrm{K}$? What is the temperature $T_R$ of the insulated surface?

### [Ex7] Radiative transfers in a semi-circular air heater

Consider an air heater consisting of a semi-circular tube for which the plane surface is maintained at $T_1 = 1\,000\,\mathrm{K}$ and the other surface of temperature $T_2$ is well insulated (Figure 11). Both surfaces are gray and Lambertian (perfect diffusers). The tube radius $r_0$ equals 20 mm, and both surfaces have an emissivity $\epsilon_1 = \epsilon_2 = 0.8$. Steady-state conditions are assumed here, and the flow is supposed to be fully developed (*i.e.* mean velocity profiles are independent of the position along the tube). Furthermore, tube end effects and axial variations in gas temperature are neglected here.

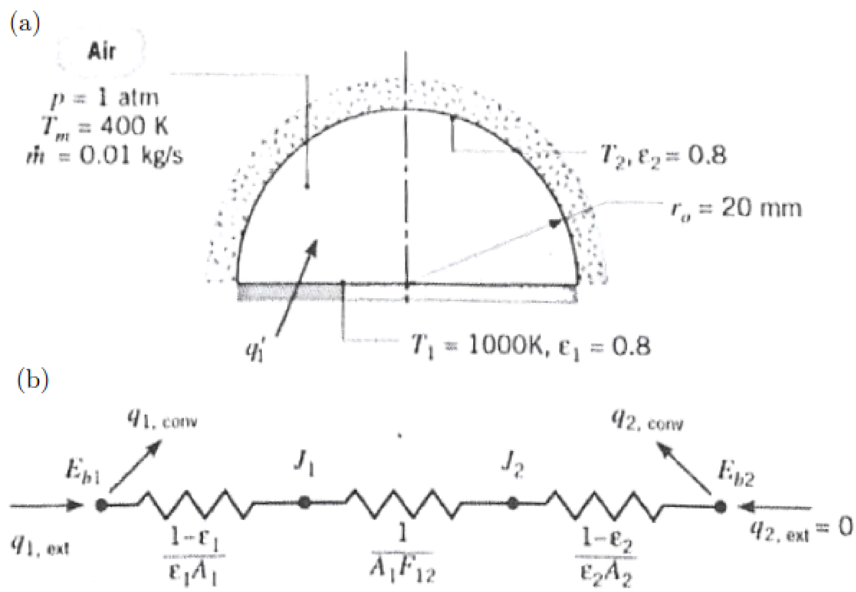


Figure 11 Schematic of the semi-circular air heater (a) and involved radiative exchanges (b).

If atmospheric air flows through the tube at $\dot{m} = 0.01\,\mathrm{kg.s^{-1}}$ and $T_m = 400\,\mathrm{K}$, what is the rate at which heat must be supplied, per unit length, to maintain the plane surface at $1\,000\,\mathrm{K}$? What is the temperature $T_2$ of the insulated surface?

The dynamic viscosity $\mu = 230.10^{-7}\,\mathrm{kg.m^{-1}.s^{-1}}$ and Prandtl number $P_r = 0.69$ for the air flow are additionally given here.